

The background features abstract green geometric shapes. On the left, a single green triangle points downwards. On the right, a complex arrangement of overlapping green triangles and polygons in various shades of green (from light lime to dark forest green) creates a layered, architectural effect. A thin white line extends from the bottom left towards the right side of the composition.

DISCRETE STRUCTURES

RELATIONS

WHY STUDY RELATIONS ?

We study relations because they help us understand and model **connections between things** in mathematics, computer science, and real life.

▶ In Mathematics:

Relations help us study connections between numbers.

For example: One number **divides** another (like 2 divides 6).

▶ In Computers:

Relations describe connections such as:

- A **program** and the **variables** it uses.
- A **programming language** and its **valid statements**.
- A **database** storing related information.

▶ In Real-World Problems:

Relations help us solve practical problems, such as:

- Finding which **cities are connected by flights**.
- Deciding the correct **order of tasks in a project**.

DOMAIN & RANGE OF RELATIONS

DOMAIN OF A RELATION:

The domain of a relation R from A to B is the set of **all first elements of the ordered pairs** which belong to R denoted $\text{Dom}(R)$.

Symbolically:

$$\text{Dom}(R) = \{a \in A \mid (a,b) \in R\}$$

RANGE OF A RELATION:

The range of A relation R from A to B is the set **of all second elements of the ordered pairs** which belong to R denoted $\text{Ran}(R)$.

Symbolically:

$$\text{Ran}(R) = \{b \in B \mid (a,b) \in R\}$$

If: $R = \{(1,2), (1,3), (2,3)\}$

Domain: All first elements, $\text{Dom}(R) = \{1,2\}$

Range: All second elements, $\text{Ran}(R) = \{2,3\}$

EXAMPLE

Let $A = \{1, 2\}$, $B = \{1, 2, 3\}$, Define a binary relation R from A to B as follows: $R = \{(a, b) \in A \times B \mid a < b\}$

Then

- Find the ordered pairs in R .
- Find the Domain and Range of R .
- Is $1R3$, $2R2$?

SOLUTION:

Given $A = \{1, 2\}$, $B = \{1, 2, 3\}$, then

$$A \times B = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}$$

a) $R = \{(a, b) \in A \times B \mid a < b\}$

$$R = \{(1,2), (1,3), (2,3)\}$$

b) $\text{Dom}(R) = \{1,2\}$ and $\text{Ran}(R) = \{2, 3\}$

c) Since $(1,3) \in R$ so $1R3$, But $(2, 2) \notin R$ so 2 is not related.

PROPERTIES OF RELATIONS

- ▶ There are several properties that are used to classify relations on a set.
- ▶ A relation R on a set A is called
 1. **reflexive** if for all $a \in A$, aRa .
Every element is related to itself.
 2. **symmetric** if for all $a, b \in A$, aRb implies bRa .
If $(a,b) \in R$ then (b,a) must also be in R
 3. **antisymmetric** if for all $a, b \in A$, aRb and bRa implies $a = b$.
If: $(a,b) \in R$ and $(b,a) \in R$ Then: $a = b$
 4. **transitive** if for all $a, b, c \in A$, aRb and bRc implies aRc .
If: $(a,b) \in R$ and $(b,c) \in R$ Then: (a,c) must also be in R

REFLEXIVE RELATION

A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$.

$$(a,a) \in R$$

Every element is related to itself.

Example $A = \{1,2,3\}$

If $R = \{(1,1),(2,2),(3,3),(1,2)\}$

This is reflexive because:

$(1,1)$

$(2,2)$

$(3,3)$

All elements relate to themselves.

If one is missing?

If $(3,3)$ is missing $\rightarrow$ NOT reflexive.

REFLEXIVE RELATION

EXAMPLE G

Consider the following relations on $\{1, 2, 3, 4\}$:

- ▶ $R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$
- ▶ $R2 = \{(1, 1), (1, 2), (2, 1)\},$
- ▶ $R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$
- ▶ $R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$
- ▶ $R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$
- ▶ $R6 = \{(3, 4)\}.$

Which of these relations are reflexive?

SOLUTION:

- ▶ The relations **R3 and R5** are **reflexive** because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$.
- ▶ The other relations are **not reflexive** because they do not contain all of these ordered pairs. In particular, **R1, R2, R4, and R6** are not reflexive because $(3, 3)$ is not in any of these relations.

SYMMETRIC & ANTISYMMETRIC RELATION

A relation R on a set A is called *symmetric* if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

AND

A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called *antisymmetric*.

- A **symmetric** relation:

A relation R on a set A is symmetric if: $(a,b) \in R \Rightarrow (b,a) \in R$

Meaning: Whenever a is related to b , b must also be related to a .

- Let $A = \{1, 2, 3\}$ so $R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ So R is symmetric

- An **antisymmetric** relation:

A relation R on A is antisymmetric if: $(a,b) \in R$ and $(b,a) \in R \Rightarrow a = b$

Meaning: If both directions happen, then a and b must be the same element.

- Let $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$

Notice: $(1, 2)$ is there $(2, 1)$ is NOT there

So no two different numbers point to each other.

EXAMPLE I

Consider the following relations on $\{1, 2, 3, 4\}$:

- ▶ $R2 = \{(1, 1), (1, 2), (2, 1)\}$,
- ▶ $R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,
- ▶ $R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,
- ▶ $R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$,
- ▶ $R6 = \{(3, 4)\}$.

Which of these relations are Symmetric and antisymmetric ?

SOLUTION:

- ▶ **The relations $R2$ and $R3$ are symmetric**, because in each case (b, a) belongs to the relation whenever (a, b) does. For $R2$, the only thing to check is that both $(2, 1)$ and $(1, 2)$ are in the relation. For $R3$, it is necessary to check that both $(1, 2)$ and $(2, 1)$ belong to the relation, and $(1, 4)$ and $(4, 1)$ belong to the relation.
- ▶ **$R4$, $R5$, and $R6$ are all antisymmetric**. For each of these relations there is no pair of elements a and b with $a \neq b$ such that both (a, b) and (b, a) belong to the relation. This is done by finding a pair (a, b) with $a \neq b$ such that (a, b) and (b, a) are both in the relation.

TRANSITIVE RELATION

A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

- If two-step paths exist, the shortcut must exist.
If no two-step path exists, nothing can go wrong.

NOTE:

Transitivity only requires that when a chain exists, the resulting pair (a, c) is in the relation.

If no chain exists, there's nothing to check $\rightarrow$ automatically OK

TRANSITIVE RELATION

A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

EXAMPLE K

Consider the following relations on $\{1, 2, 3, 4\}$:

- ▶ $R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$,
- ▶ $R2 = \{(1, 1), (1, 2), (2, 1)\}$,
- ▶ $R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,
- ▶ $R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,
- ▶ $R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$.

Which of these relations are transitive ?

SOLUTION:

$R4$ and $R5$ are transitive. For each of these relations, we can show that it is transitive by verifying that if (a, b) and (b, c) belong to this relation, then (a, c) also does. For instance, $R4$ is transitive, because of $(3, 2)$ and $(2, 1)$, $(4, 2)$ and $(2, 1)$, $(4, 3)$ and $(3, 1)$, and $(4, 3)$ and $(3, 2)$ are the only such sets of pairs, and $(3, 1)$, $(4, 1)$, and $(4, 2)$ belong to $R4$.

$R1$ is not transitive because $(3, 4)$ and $(4, 1)$ belong to $R1$, but $(3, 1)$ does not.

$R2$ is not transitive because $(2, 1)$ and $(1, 2)$ belong to $R2$, but $(2, 2)$ does not.

$R3$ is not transitive because $(4, 1)$ and $(1, 2)$ belong to $R3$, but $(4, 2)$ does not.

Transitivity means: ALL possible chains must satisfy the rule.

METHODS FOR REPRESENTING RELATIONS

► Two methods for representing relations:

1. **Square Matrices**
2. **Directed Graphs** (consisting of vertices and directed edges)

REPRESENTING RELATIONS USING MATRICES

A relation between finite sets can be shown using a matrix of 0s and 1s.

Suppose: $A = \{a_1, a_2, \dots, a_m\}$ & $B = \{b_1, b_2, \dots, b_n\}$

▶ The relation R can be represented by the matrix $M_R = [m_{ij}]$, where

- Rows $\rightarrow$ elements of A
- Columns $\rightarrow$ elements of B

For each position (i, j) :

- Put 1 if a_i is related to b_j
- Put 0 if a_i is not related to b_j

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

- Consider the relation of who is enrolled in which class
 - Let $A = \{ \text{Alice, Bob, Claire, Dan} \}$
 - Let $B = \{ \text{CS101, CS201, CS202} \}$
 - $R = \{ (a,b) \mid \text{person } a \text{ is enrolled in course } b \}$

	CS101	CS201	CS202
Alice	X		
Bob		X	X
Claire			
Dan		X	X

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Examples of Representing Relations Using Matrices (*cont.*)

Example 3: Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which ordered pairs are in the relation R represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} ?$$

Solution:

$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}$.

Matrices of Relations on Sets

If R is a reflexive relation, all the elements on the main diagonal of M_R are equal to 1.

$$\mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrices of Relations on Sets

R is a symmetric relation

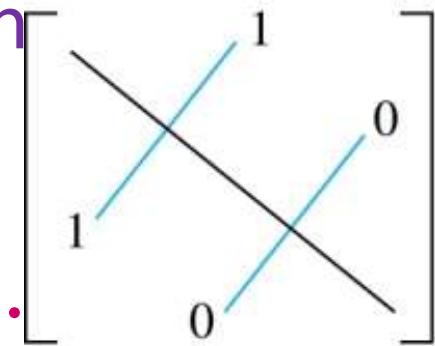
If a_i is related to a_j , then a_j is related to a_i .

In the matrix:

If position (i, j) is 1, then position (j, i) must also be 1.

So the matrix looks mirrored across the

R is a symmetric relation,
if and only if $m_{ij} = 1$ whenever $m_{ji} = 1$.



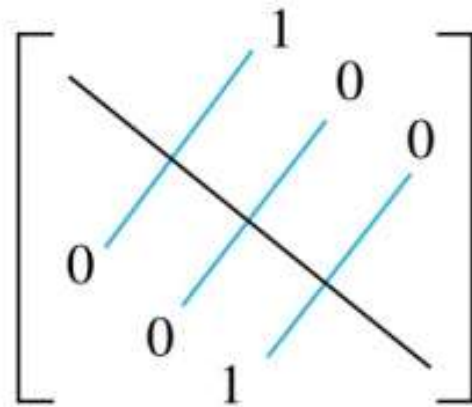
(a) Symmetric

Matrices of Relations on Sets

The matrix of an antisymmetric relation has the property that if $m_{ij}=1$ with $i \neq j$, then $m_{ji}=0$
OR Either $m_{ij}=0$ or $m_{ji}=0$ when $i \neq j$

In other words:

- If $i \neq j$, at least one of the two entries must be 0.



(b) Antisymmetric

Matrices of Relations on Sets

Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

Solution:

Because all the diagonal elements are equal to 1, R is reflexive.

Because M_R is symmetric, R is symmetric and not antisymmetric because both $m_{1,2}$ and $m_{2,1}$ are 1.

REPRESENTING RELATIONS USING Directed Graph (Digraph)

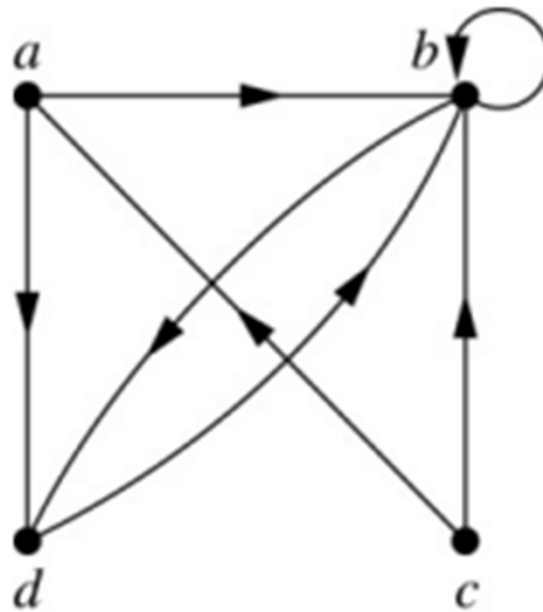
- A **directed graph** (or **digraph**) has:
 - A set of **vertices** (points or nodes) called **V**.
 - A set of **edges** (arrows) called **E**.
- Each edge is an **ordered pair** like (a, b) , meaning an arrow from **a** to **b**.
- The vertex **a** is the **start** (initial vertex) of the arrow.
- The vertex **b** is the **end** (terminal vertex) of the arrow.
- If an edge goes from a vertex back to itself, like (a, a) , it's called a **loop**.

In short:

- Vertices = points
- Edges = arrows between points
- Loop = arrow that starts and ends at the same point

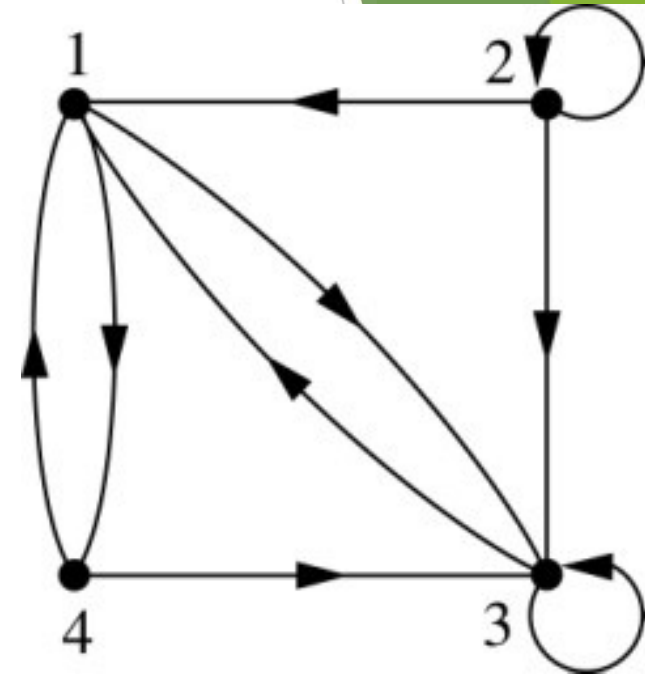
REPRESENTING RELATIONS USING Directed Graph (Digraph)

- A drawing of the directed graph with
- vertices a , b , c , and d , and
- edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , (d, b) is shown here.



REPRESENTING RELATIONS USING Directed Graph (Digraph)

Example : What are the ordered pairs in the relation represented by this directed graph?



Solution: The ordered pairs in the relation are

$(1, 3)$, $(1, 4)$, $(2, 1)$, $(2, 2)$, $(2, 3)$, $(3, 1)$, $(3, 3)$, $(4, 1)$, and $(4, 3)$

Determining which Properties a Relation has from its Digraph

- ▶ **1. Reflexive:** Every vertex must have a loop.
- ▶ That means:
- ▶ For every element x
- ▶ The pair (x, x) must exist
- ▶ In the graph: Every vertex must have a small circle (loop) on itself.
- ▶ If even one vertex does not have a loop, Then it is NOT reflexive.

- ▶ **2. Symmetric**
- ▶ If there is an arrow from x to y , there must also be an arrow from y to x .
- ▶ So If (x, y) exists, then (y, x) must also exist.
- ▶ In the graph: Every arrow must have a matching arrow going back.

3. Antisymmetric

- If $x \rightarrow y$ exists (and $x \neq y$), then $y \rightarrow x$ must NOT exist.
- So you cannot have arrows both ways between two different vertices.
- If you see: $x \rightarrow y$
and $y \rightarrow x$
and $x \neq y$, then it is NOT antisymmetric.

4. Transitive: This means:

If you can go from $x \rightarrow y$
and from $y \rightarrow z$
then you must also be able to go from $x \rightarrow z$.

So if: $x \rightarrow y$
 $y \rightarrow z$
Then $(x \rightarrow z)$ must also be present.
If that direct arrow is missing
Then it is NOT transitive.

DIRECTED GRAPH OF A REFLEXIVE RELATION

The directed graph of every reflexive relation includes an arrow from every point to the point itself (i.e., a loop).

EXAMPLE Q

Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 , R_3 , and R_4 on A by

$$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$$

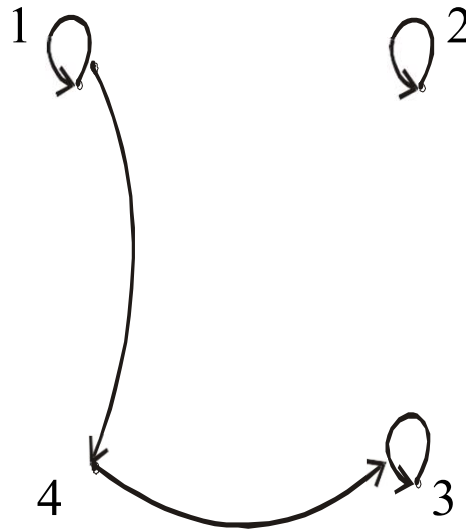
$$R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_4 = \{(1, 4), (2, 2), (2, 3), (4, 1), (3, 3)\}$$

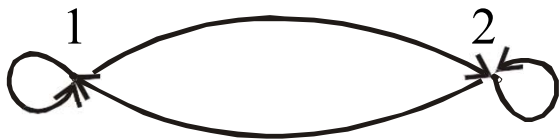
Then their directed graphs are:



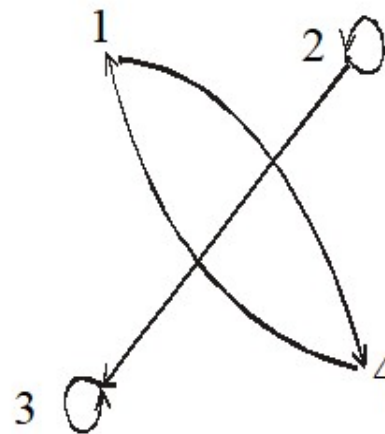
R_1 is reflexive because at every point of the set A we have a loop in the graph.



R_2 is not reflexive, as there is no loop at 4.



R_3 is reflexive



R_4 is not reflexive, as there are no loops at 1 and 4.

GRAPH OF A SYMMETRIC RELATION

For a symmetric directed graph whenever there is an arrow going from one point of the graph to a second, there is an arrow going from the second point back to the first.

EXAMPLE S

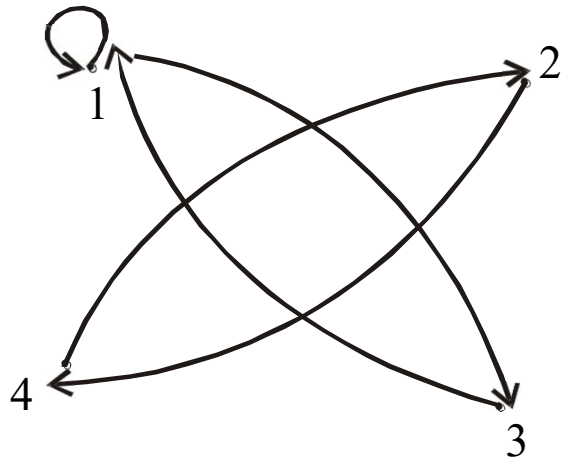
Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 , R_3 , and R_4 on A by the directed graphs:

$$R_1 = \{(1, 1), (1, 3), (2, 4), (3, 1), (4, 2)\}$$

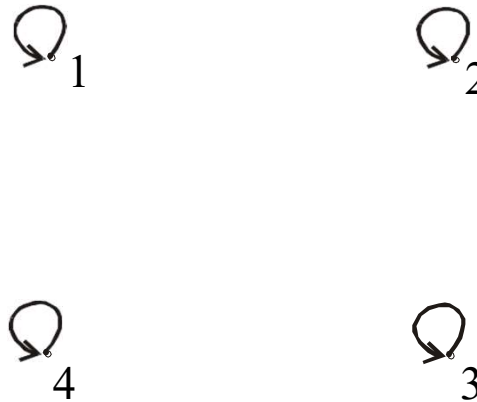
$$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_3 = \{(2, 2), (2, 3), (3, 4)\}$$

$$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$$

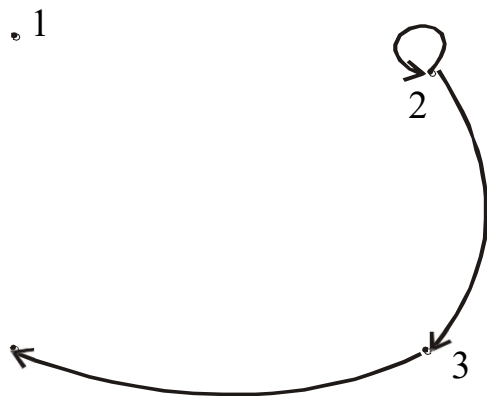


R_1 is symmetric



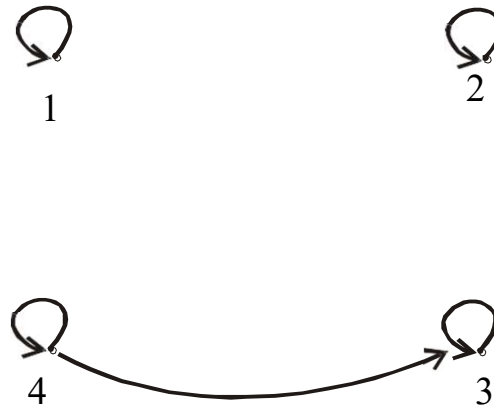
R_2 is symmetric

Because reversing any pair gives the same pair, which is already in the relation.



R_3 is not symmetric since there are arrows from 2 to 3 and from 3 to 4 but not conversely

But R_3 is antisymmetric.



R_4 is not symmetric since there is an arrow from 4 to 3 but no arrow from 3 to 4

But R_4 is antisymmetric.

DIRECTED GRAPH OF A TRANSITIVE RELATION

For a transitive directed graph, whenever there is an arrow going from one point to the second, and from the second to the third, there is an arrow going directly from the first to the third.

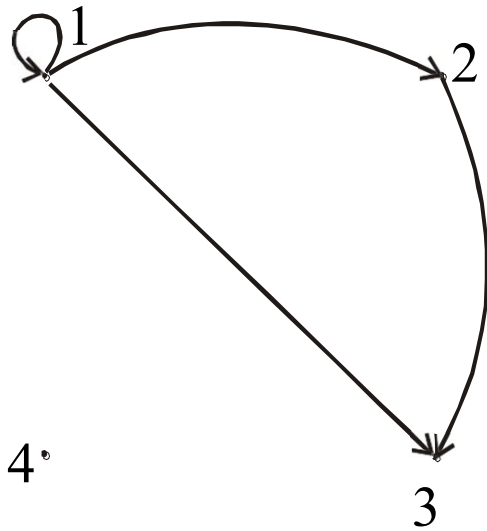
EXAMPLE W

Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 and R_3 on A by the directed graphs:

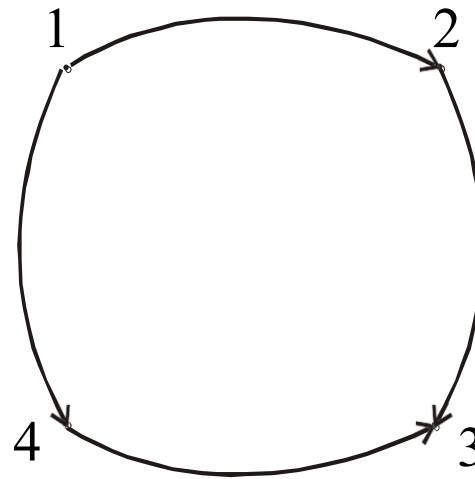
$$R_1 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$$

$$R_2 = \{(1, 2), (1, 4), (2, 3), (4, 3)\}$$

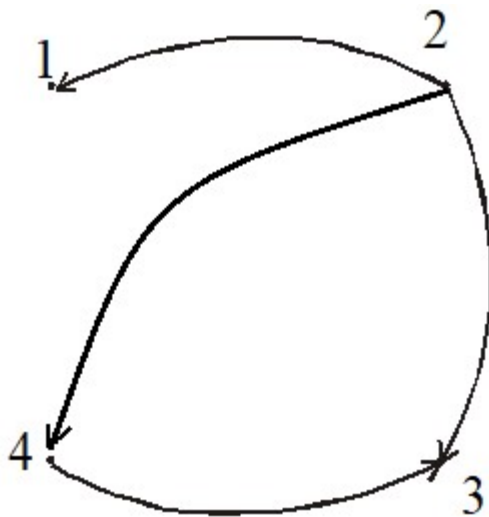
$$R_3 = \{(2, 1), (2, 4), (2, 3), (4, 3)\}$$



R_1 is transitive

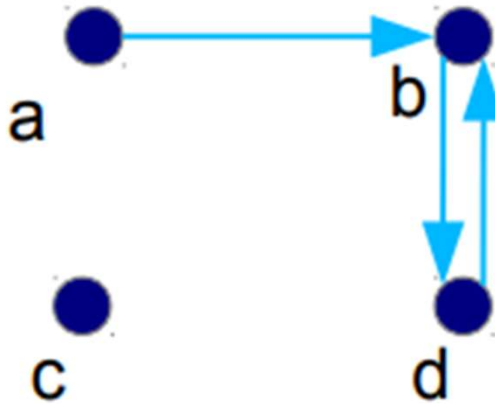


R_2 is not transitive since there is an arrow from 1 to 2 and from 2 to 3 but no arrow from 1 to 3 directly



R_3 is transitive

Determining which Properties a Relation has from its Digraph



- Reflexive? No, there are no loops
- Symmetric? No, there is an edge from a to b, but not from b to a
- Antisymmetric? No, there is an edge from d to b and b to d
- Transitive? No, there are edges from a to b and from b to d, but there is no edge from a to d

Composition

Composition means **linking two relations through a common middle element.**

- Let R_1 connect set **A** to set **B**.
- Let R_2 connect set **B** to set **C**.

To form the composition $R_2 \circ R_1$ (read: “ R_2 after R_1 ”):

- If R_1 connects **x** to **y**,
- And R_2 connects **y** to **z**,
- Then the composition connects **x** to **z**.
- **Match the middle element (y), & connect the first to the last.**

Let the sets be:

- $A = \{1, 2\}$, $B = \{a, b\}$, $C = \{X, Y\}$
- $R_1 = \{ (1, a), (2, b) \}$
- $R_2 = \{ (a, X), (b, Y) \}$
- $(R_2 \circ R_1) = \{ (1, X), (2, Y) \}$

Composition

Let

$$A = \{p, q\}$$

$$B = \{1, 2\}$$

$$C = \{\alpha, \beta\}$$

SOLUTION:

Relations:

$$R_1 = \{(p, 1), (q, 2)\}$$

$$R_2 = \{(1, \alpha), (2, \beta)\}$$

Composition $R_2 \circ R_1$:

p connects to 1 via R_1 , and 1 connects to α via $R_2 \rightarrow$ connect p to α

q connects to 2 via R_1 , and 2 connects to β via $R_2 \rightarrow$ connect q to β

So,

$$R_2 \circ R_1 = \{(p, \alpha), (q, \beta)\}$$

Power of a Relation

Powers of a relation mean applying the same relation again and again.

Let R be a binary relation on A . Then the powers R^n of the relation R can be defined inductively by:

- Basis Step: $R^1 = R$
 - ▶ (The first power is just the relation itself)
 - Inductive Step: $R^{n+1} = R^n \circ R$ *means*
 - ▶ $R^2 = R \circ R$
 - ▶ $R^3 = R^2 \circ R$
 $R^4 = R^3 \circ R$
- ▶ Each time:
 - ▶ Take previous result
 - ▶ Apply relation again

Power-Example

Given Relation $R = \{(a,b), (b,c), (c,d)\}$

Step 1: $R^1 = R = \{(a,b), (b,c), (c,d)\}$

Step 2: $R^2 = R \circ R$

Take previous result & Apply relation again

Now check all possible chains for R^2

(a,b) and $(b,c) \rightarrow (a,c)$

(b,c) and $(c,d) \rightarrow (b,d)$

So: $R^2 = \{(a,c), (b,d)\}$

► Step 3: $R^3 = R^2 \circ R$ (Now combine R^2 with R)

► From (a,c) : (c,d) exists $\rightarrow (a,d)$

► From (b,d) : No pair starts with $d \rightarrow$ nothing

► So: $R^3 = \{(a,d)\}$

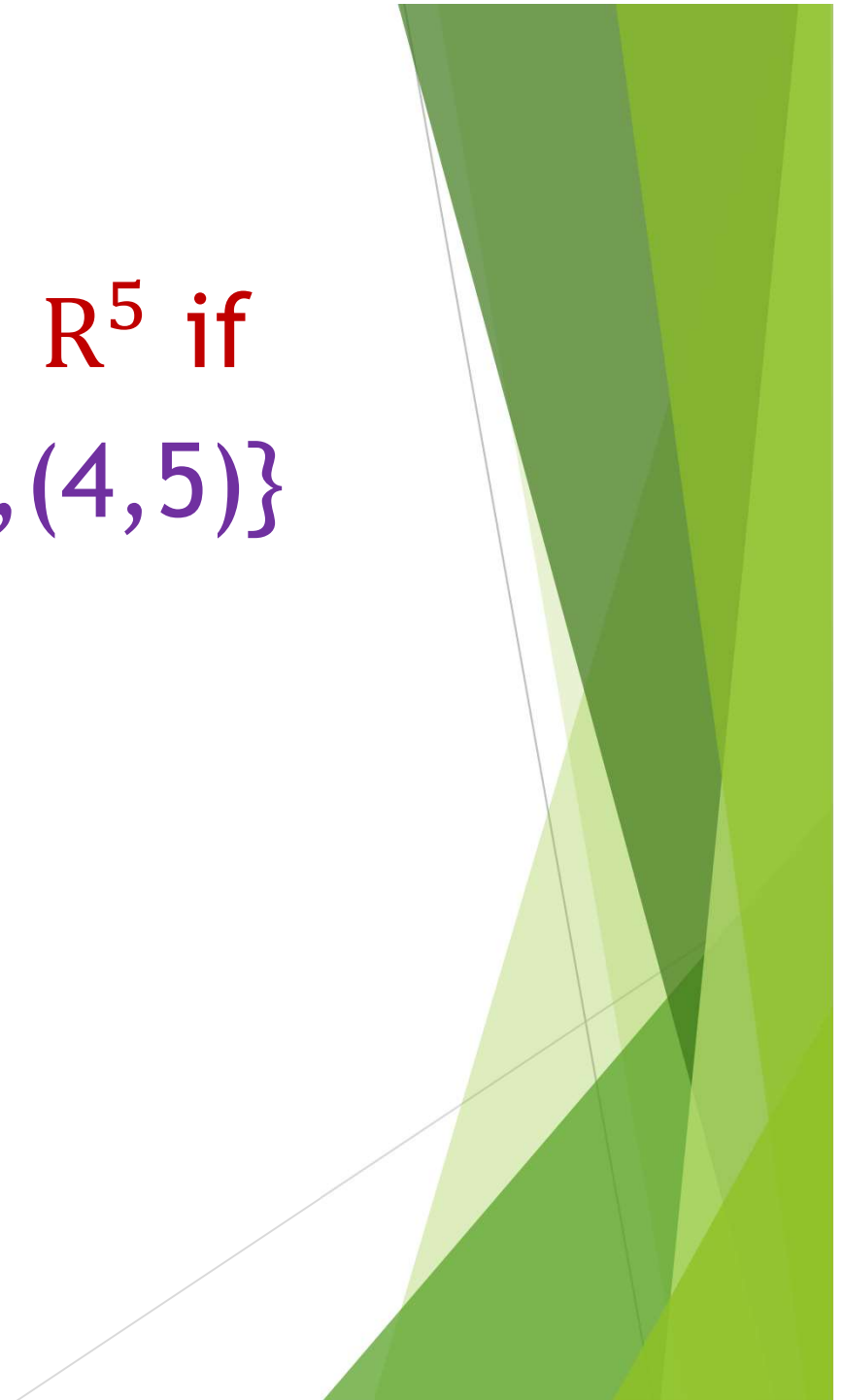
► Step 4: $R^4 = R^3 \circ R$

► From (a,d) : No $(d, ?)$ in $R \rightarrow$ nothing

► So: $R^4 = \emptyset$

Power-Example

► Find R , R^2 , R^3 , R^4 , R^5 if
 $R = \{(1,2), (2,3), (3,4), (4,5)\}$



Step 1: $R^1 = \{(1,2), (2,3), (3,4), (4,5)\}$

► **Step 2:** $R^2 = R \circ R$

► $(1,2)$ and $(2,3) \rightarrow (1,3)$

► $(2,3)$ and $(3,4) \rightarrow (2,4)$

► $(3,4)$ and $(4,5) \rightarrow (3,5)$

► $R^2 = \{(1,3), (2,4), (3,5)\}$

► **Step 3:** $R^3 = R^2 \circ R$

► Combine R^2 with R

► $(1,3)$ and $(3,4) \rightarrow (1,4)$

► $(2,4)$ and $(4,5) \rightarrow (2,5)$

► $(3,5) \rightarrow$ no pair starts with 5 in $R \rightarrow$ nothing

► $R^3 = \{(1,4), (2,5)\}$

► **Step 4:** $R^4 = R^3 \circ R$

► $(1,4)$ and $(4,5) \rightarrow (1,5)$

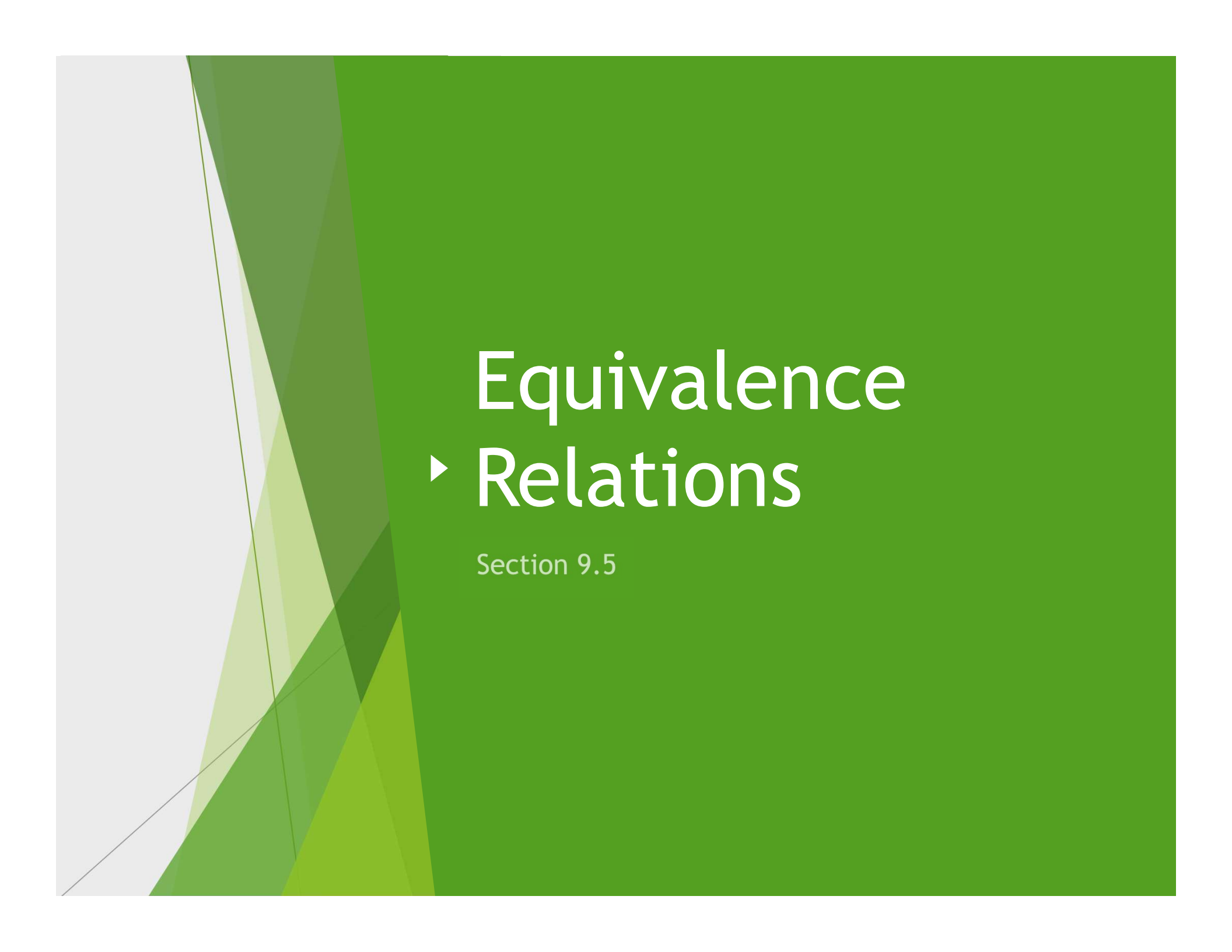
► $(2,5) \rightarrow$ no pair starts with 5 in $R \rightarrow$ nothing

► $R^4 = \{(1,5)\}$

► **Step 5:** $R^5 = R^4 \circ R$

► $(1,5) \rightarrow$ no pair starts with 5 in $R \rightarrow$ nothing

► $R^5 = \emptyset$



Equivalence

▸ Relations

Section 9.5

Equivalence Relations

- ▶ A relation is called an **equivalence relation** if it has all these **3 properties**:
 1. **Reflexive** $\rightarrow$ every element relates to itself
 2. **Symmetric** $\rightarrow$ if a relates to b , then b relates to a
 3. **Transitive** $\rightarrow$ if $a \rightarrow b$ and $b \rightarrow c$, then $a \rightarrow c$
- ▶ The notation $a \sim b$ is used to denote that a and b are equivalent

Strings (Length of words)

► **Equivalence:** Two strings are related if they have same length

► **Example:**

1. "cat" and "dog" $\rightarrow$ related (length 3)
2. "hi" and "no" $\rightarrow$ related (length 2)
3. "abc" $\sim$ "xyz" $\sim$ "def" $\rightarrow$ related (length 3)

► **Check properties:**

► ✓ Reflexive:

Every string has same length as itself

► ✓ Symmetric:

If a and b same length $\rightarrow$ b and a also same

► ✓ Transitive:

If a and b same length, b and c same $\rightarrow$ a and c same

Congruence Modulo m

- ▶ Rule: $a \equiv b \pmod{m}$
- ▶ Means: a & b are equivalent when both give same remainder when divided by m

- ▶ Example 1

- ▶ $7 \equiv 4 \pmod{3}$
Both remainder = 1

- ▶ Example 2:

- ▶ $12 \equiv 7 \pmod{5}$
Both remainder = 2

- ▶ Example 3

- ▶ $10 \equiv 1 \pmod{3}$
Both remainder = 1

Check properties:

✓ Reflexive:

$$a \equiv a$$

✓ Symmetric:

$$\text{if } a \equiv b \rightarrow b \equiv a$$

✓ Transitive:

$$\text{if } a \equiv b \text{ and } b \equiv c \rightarrow a \equiv c$$

Divides

► Example

- Show that the “divides” relation on positive integers is **not an equivalence relation**.

► Solution

- To be an equivalence relation, a relation must be **reflexive, symmetric, and transitive**.

- **Reflexive:**

Every number divides itself, So, a/a ✓

- **Symmetric (fails):**

If a/b , it does **not** always mean b/a ✗

- **Transitive:** If a/b and b/c , then a/c ✓

Take numbers: 2, 4, and 8

- **Reflexive:**

$2/2, 4/4, 8/8$

→ Every number divides itself

- **Not Symmetric:**

2 divides 4

But 4 does NOT divide 2

→ So symmetry fails

- **Transitive:**

$1/2 = 2/4$

$2/4 = 4/8$

⇒ So: $1/2 = 4/8$

Equivalence Classes

- ▶ An **equivalence class** is:
- ▶ A group of elements that are **related (similar)** to each other.
- ▶ **Purpose of Equivalence Classes**
 - They help us **group things that are essentially the same**.
 - Instead of treating every element separately, we treat **similar ones as one group**.
- ▶ If you have a relation R on a set:
- ▶ The equivalence class of a , written as $[a]$, is all elements that are related to a
- ▶ So: rep. by
- ▶ $[a] = \{all\ elements\ related\ to\ a\}$
- ▶ **Example (mod 4)**
- ▶ Numbers: 1, 5, 9, 13
 - All give **remainder 1** when divided by 4
 - So instead of writing all of them, we just say: $[1]_4$

Equivalence Classes

▶ **Group 1: $[0]_4$ (Numbers with remainder 0)**

▶ $[0]_4 = \{..., -8, -4, 0, 4, 8, ...\}$

▶ **Group 2: $[1]_4$ (Numbers with remainder 1)**

▶ $[1]_4 = \{..., -7, -3, 1, 5, 9, ...\}$

▶ **Group 3: $[2]_4$**

▶ $[2]_4 = \{..., -6, -2, 2, 6, 10, ...\}$

▶ **Group 4: $[3]_4$**

▶ $[3]_4 = \{..., -5, -1, 3, 7, 11, ...\}$

Equivalence Classes

► Example: Mod 3 Groups

- Define equivalence classes modulo 3:
- Group 1: $[0]_3 = \{\dots, -6, -3, 0, 3, 6, \dots\}$
- Group 2: $[1]_3 = \{\dots, -5, -2, 1, 4, 7, \dots\}$
- Group 3: $[2]_3 = \{\dots, -4, -1, 2, 5, 8, \dots\}$

► Example: Mod 5 Groups

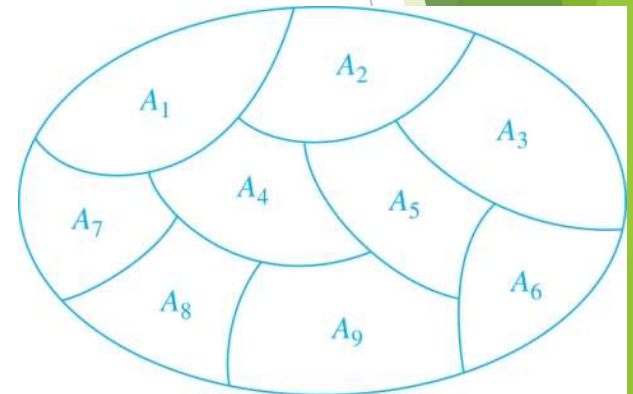
- Define equivalence classes modulo 5:
- Group 1: $[0]_5 = \{\dots, -10, -5, 0, 5, 10, \dots\}$
- Group 2: $[1]_5 = \{\dots, -9, -4, 1, 6, 11, \dots\}$
- Group 3: $[2]_5 = \{\dots, -8, -3, 2, 7, 12, \dots\}$
- Group 4: $[3]_5 = \{\dots, -7, -2, 3, 8, 13, \dots\}$
- Group 5: $[4]_5 = \{\dots, -6, -1, 4, 9, 14, \dots\}$

Partition of a Set

- ▶ A partition of a set S means:
 - You break S into smaller groups (subsets)
 - These groups don't overlap (no shared elements)
 - None of the groups are empty
 - If you put all groups together, you get the entire set S

- ▶ Conditions:

- ▶ $A_i \neq \emptyset$:
 - Each group has at least one element
- ▶ $A_i \cap A_j = \emptyset$ when $i \neq j$:
 - Different groups don't share elements
- ▶ $\bigcup_{i \in I} A_i = S$:
 - Combining all groups gives back the full set S



Equivalence Relation Creates a Partition

- If you have an equivalence relation R on a set A , it splits A into groups called **equivalence classes**.
- Every element a is in its own class $[a]_R$
- These classes **don't overlap** (if two classes share an element, they are the same class).
- All classes together cover the entire set A .
- ▶ **So, equivalence classes form a partition of A .**

An Equivalence Relation- Partitions a Set

- ▶ Let R be an equivalence relation on a set A . The union of all the equivalence classes of R is all of A , since an element a of A is in its own equivalence class $[a]_R$. In other words,

$$\bigcup_{a \in A} [a]_R = A.$$

- ▶ **From Theorem 1,**
- ▶ **Equivalence classes are either the same or completely separate.**
 - ▶ If $[a]_R \neq [b]_R$, then $[a]_R \cap [b]_R = \emptyset$.
- ▶ This means the equivalence classes **divide the set A into non-overlapping groups**, **forming a partition of A .**
- ▶ So, Each element belongs to exactly one group, and the groups together cover the whole set.

An Equivalence Relation Partitions a Set

Theorem 2:

•Equivalence relation $\rightarrow$ partition:

If R is an equivalence relation on a set S , then its equivalence classes **split S into separate groups** (a partition).

•Partition $\rightarrow$ equivalence relation:

If you have a partition of S (groups that don't overlap and cover the set), you can define a relation R where **two elements are related if they are in the same group**. This relation is an equivalence relation.

Proof Idea

Step 1: Reflexive

Every element is in the same group as itself $\rightarrow (a,a) \in R$.

Step 2: Symmetric

If a and b are in the same group $\rightarrow b$ and a are also in the same group $\rightarrow (b,a) \in R$.

Step 3: Transitive

If a and b are in the same group, and b and c are in the same group $\rightarrow$ all three must be in the same group $\rightarrow (a,c) \in R$.

So, the relation satisfies **reflexive, symmetric, transitive**, which makes it an **equivalence relation**.

An Equivalence Relation

Partitions a Set

Let $S = \{1, 2, 3, 4, 5, 6\}$

Equivalence class: “numbers have the same remainder when divided by 2”

Step 1: Make groups (partition)

- Even numbers $\rightarrow \{2, 4, 6\}$
- Odd numbers $\rightarrow \{1, 3, 5\}$

These are **non-overlapping groups** and cover all elements. So, this is a **partition**

Step 2: Make relation from groups {Every group element can relate itself}

“a is related to b if they are in the same group”

Examples:

- $2 \sim 4$ (both even)
- $1 \sim 5$ (both odd)
- $2 \not\sim 3$ (different groups)

Step 3: Check properties

- Reflexive: $2 \sim 2$ (2 is even, and 2 is even $\rightarrow$ same type)
- Symmetric: if $2 \sim 4 \rightarrow 4 \sim 2$ (2 and 4 are both even $\rightarrow$ same group, order doesn't matter)
- Transitive: if $2 \sim 4$ and $4 \sim 6 \rightarrow 2 \sim 6$ (2, 4, 6 are all even, so 2 and 6 are also related)



► Partial Orderings

Section 9.6

Partial Ordering

A **partial order** is a way to organize things when:

- **some elements can be compared & some cannot**
- **JUST LIKE**

Total Order (everything comparable):

- Heights of students → $170\text{cm} > 160\text{cm}$
- Numbers → $5 > 3$
- Ages → $20 > 18$

Partial Order (not always comparable):

- Skills → Math vs English
- Subjects difficulty → Programming vs History
- Tasks → “Study” vs “Exercise” (no fixed order)

A partial order is just a relation that follows 3 rules:

- ✓ Reflexive
- ✓ Antisymmetric
- ✓ Transitive

If a relation has these 3 properties, we call it a **partial ordering**

Partial Ordering

Example 1: Show that the “greater than or equal” relation ($\geq$) is a partial ordering on the set of integers.

1. *Reflexivity:* $a \geq a$ for every integer a .
2. *Antisymmetry:* If $a \geq b$ and $b \geq a$, then $a = b$.
3. *Transitivity:* If $a \geq b$ and $b \geq c$, then $a \geq c$.

Partial Ordering

1. Reflexive-“Self-related”

Numbers:

- $8 \geq 8$ ✓
- $-3 \leq -3$ ✓

Sets:

- $\{a\} \subseteq \{a\}$ ✓

2. Antisymmetric-“No two different elements dominate each other”

$$aRb \text{ and } bRa \Rightarrow a = b$$

Numbers

- $9 \geq 4$ ✓ but $4 \geq 9$ ✗ Not both $\rightarrow$ OK
- $6 \geq 6$, $6 \geq 6$ ✓ as both equal

Subsets

- $\{1\} \subseteq \{1, 2\}$ ✓
- $\{1, 2\} \subseteq \{1\}$ ✗ Not both $\rightarrow$ OK
- $A \subseteq B$ and $B \subseteq A$ then $A = B$ ✓ equal

3. Transitive – “Chain works” $aRb \text{ and } bRc \Rightarrow aRc$

Numbers

- $15 > 10$, $10 > 5$
so, $15 > 5$ ✓

Subsets

- $\{a\} \subseteq \{a, b\} \subseteq \{a, b, c\}$
so, $\{a\} \subseteq \{a, b, c\}$ ✓

Partial Orderings (*continued*)

Example 2: Show that divisibility ($|$) is a partial order on positive integers $\mathbb{Z}^+$

- ▶ **Reflexivity:** $a | a$ for all integers a .
- ▶ **Antisymmetry:** If a and b are positive integers with $a | b$ and $b | a$, then $a = b$.
- ▶ **Transitivity:** Suppose that a divides b and b divides c . Then $a|b$ and $b|c \Rightarrow a|c$. Therefore, the relation is transitive.

Comparability

The symbol $\leq$ is used to denote it.

- ▶ Two elements a and b are comparable if:
- ▶ either a comes before $b \rightarrow a \leq b$
- ▶ or b comes before $a \rightarrow b \leq a$
- ▶ If neither is true, they are **incomparable**.

Example 1: Numbers

Let $S = \{1, 2, 3\}$, and relation is “ $\leq$ ”.

- 1 and 2 $\rightarrow$ comparable ($1 \leq 2$)
- 3 and 2 $\rightarrow$ comparable ($2 \leq 3$)

Here, pairs is comparable

Example 2: Divisibility

Let $S = \{2, 3, 6\}$, relation = “divides” ($|$)

- $2 \mid 6 \rightarrow$ comparable
- $3 \mid 6 \rightarrow$ comparable
- BUT:

- 2 does NOT divide 3
- 3 does NOT divide 2

So 2 and 3 are incomparable

This is the key idea:

Not all elements can always be compared!

It's a pair property

Totally Ordered Set

- ▶ A set is **totally ordered** if:
 - ▶ You can compare every pair of elements
- ▶ For any a, b :
- ▶ either $a \leq b$
- ▶ or $b \leq a$
- ▶ “Order” sounds like: everything should be arranged in a line
- ▶ But the **real condition** is:
 - ▶ No two elements are left **uncomparable**

Totally Ordered Set

Example 1: Numbers with $\leq$

Set: $\{1, 2, 3, 4\}$

- $1 \text{ vs } 3 \rightarrow \text{comparable}$
- $2 \text{ vs } 4 \rightarrow \text{comparable}$
- Any pair $\rightarrow \text{comparable}$

This is a **total order**

Total order = every pair can be compared

Example 2: Divisibility

Set: $\{2, 3, 6\}$, relation = “divides”

- $2 \mid 6 \rightarrow \text{OK}$
- $3 \mid 6 \rightarrow \text{OK}$
- BUT 2 and 3:
 - 2 does not divide 3
 - 3 does not divide 2

Not comparable $\rightarrow$ **NOT total order**

Well Ordered Set

A set is **well-ordered** if:

- It is totally ordered
- Every non-empty subset has a smallest element

Such as:

Set: $\mathbb{N} = \{1, 2, 3, 4, \dots\}$

Totally ordered (numbers can be compared)

Any subset has a smallest element

Example subset:

$\{5, 10, 20\} \rightarrow$ smallest is 5

$\{100, 200\} \rightarrow$ smallest is 100

So this is **well-ordered**

Example

Set: all integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

Consider its subset: $\{\dots, -3, -2, -1\}$

No smallest element (it keeps going down) So **NOT well-ordered**

Lexicographic Set

Lexicographic order is a way to arrange **words or strings** in the same way a **dictionary** does.

How it works

- Compare two words **letter by letter from left to right**
- The **first different letter decides the order**
- If one word ends first, the **shorter word comes first**

Examples

1. **apple < banana**
(a comes before b)
2. **apple < apricot**
(a = a, p = p, p < r)
3. **app < apple**
(app is part of apple → shorter comes first)
4. **car < cat**
(r comes before t)

Lexicographic Set

Important point (When numbers are treated as strings, they are compared like words:)

Numbers treated as strings are different:

Lexicographic order: "10" < "2" (because '1' < '2')

Numerical order: 2 < 10

Example: ["1", "2", "10", "20"]

Lexicographic sort:

Compare as strings:

"1"

"10"

"2"

"20"

Order becomes: [1, 10, 2, 20]

Not correct: Numerical sort [1, 2, 10, 20]

Lexicographic Set

Sort using Lexicographic Order (NUMBERS)

- "1"
- "10"
- "2"
- "20"
- "3"
- "11"
- "12"

Sort using Lexicographic Order (STRINGS)

- "zebra"
- "apple"
- "banana"
- "car"
- "apricot"
- "dog"
- "ant"
- "bat"
- "cat"
- "ball"

Hasse Diagrams

- ▶ A **Hasse diagram** is a simple picture of a partial order that removes extra unnecessary edges.
- ▶ It does NOT show:
 - ▶ self-loops (because of reflexive property)
 - ▶ indirect edges (because of transitive property)
 - ▶ It only shows the **important direct connections**.

Hasse Diagrams

- ▶ To construct a Hasse diagram, we follow these four steps:
- ▶ **Step 1**
- ▶ Create a directed graph from the relation.



Directed Graph

- ▶ Step 2
- ▶ Remove all self-loops.



Remove Loops

- Step 3
- Remove all transitivity.



Remove Transitivity

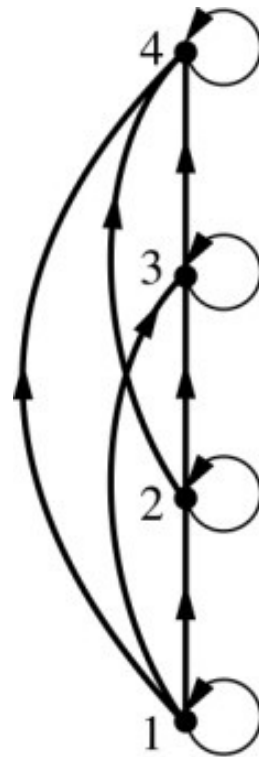
- Step 4
- Remove directed arrows.



Hasse Diagram

Remove Directed Arrows

Hasse Diagrams



(a)



(b)



(c)

STEPS

- Draw all relations
- Remove self-loops ($a \rightarrow a$)
- Remove indirect (transitive) edges
- Place smaller elements below larger ones
- Remove arrows (assume upward direction)

Hasse Diagrams

- ▶ Let $S = \{1, 2, 3, 5, 10, 15\}$
- ▶ Define the relation R on S as:
 - ▶ aRb if a divides b

Task:

- ▶ Write all ordered pairs in the relation R .
- ▶ Draw the directed graph of R .
- ▶ Remove:
 - ▶ reflexive edges (loops)
 - ▶ transitive edges
- ▶ Draw the **Hasse diagram** of (S, R) .

Hasse Diagrams

► 1. Relation R (divides)

From 1: 1 divides everything: (1,2), (1,3), (1,5), (1,10), (1,15)

From 2: (2,10)

From 3: (3,15)

From 5: (5,10), (5,15)

From 10, 15: (10,10), (15,15)

So,

$R = (1,2), (1,3), (1,5), (1,10), (1,15), (2,10), (3,15), (5,10), (5,15), (10,10), (15,15)$

Hasse Diagrams

► 2. Remove reflexive pairs (loops)

$(1,1), (2,2), (3,3), (5,5), (10,10), (15,15)$

► 3. Remove transitive edges

Remove:

$1 \rightarrow 10$ (because $1 \rightarrow 2 \rightarrow 10$)

► $1 \rightarrow 15$ (because $1 \rightarrow 3 \rightarrow 15$)

Final Hasse edges

$(1,2), (1,3), (1,5), (2,10), (3,15), (5,10), (5,15)$

► Visual Levels:

► Top: 10, 15

► Middle: 2, 3, 5

► Bottom: 1

